Document Control

- Document Title: Business Requirements Document (BRD)
- System Name: Application
- Version: 1.0
- Date: Not present in source JSON
- Author: Not present in source JSON

Executive Summary

- The Business Requirements Document details the ingestion of customer and order CSV data into Delta tables.
- It outlines data cleansing, total computation, SCD Type 2 processing, join logic, and aggregation into customeraggregatespend.
- System Name: Application

Business Objectives

- Load source CSVs into Delta tables for customer and order data.
- Cleanse data by removing "Null"/Null literals and duplicates.
- Enrich orders with a computed TotalAmount = PricePerUnit × Qty.
- Produce an ordersummary table by joining customer and order on CustId.
- Apply SCD Type 2 change handling for customer data, maintaining full history.
- Create a customeraggregatespend table aggregating spend per Name per Date.
- Execute the end-to-end pipeline using Delta Live Tables.

In-Scope

- Source ingestion from specified file system locations.
- Creation and management of Delta tables: customer, order, ordersummary, customeraggregatespend.
- Data cleansing actions as stated.
- Computation of TotalAmount in order.
- Join logic and SCD Type 2 handling for customer changes in ordersummary.
- Aggregate processing from ordersummary to customeraggregatespend.
- Implementation guidance tied to Delta Live Tables.

Out-of-Scope

- Any data sources, systems, or fields not present in source JSON.
- Security, access controls, SLAs, monitoring, and alerting.
- Performance tuning and scaling parameters.

- Scheduling, orchestration, or DevOps details beyond “Implement steps 1–10 using Delta Live Tables.”
- Data quality metrics beyond the cleansing actions listed.
- User interfaces, reports, or analytics consumption layers.

Stakeholders

- Not present in source JSON.

Business Requirements

BR-REQ-001 — Source Ingestion

- Description: Load source CSVs into Delta tables from specified paths.
- Business Value: Enables core data pipeline and foundational data access.
- Priority: High

BR-REQ-002 — Schema Definition

- Description: Define schemas for customer and order tables per specified columns.
- Business Value: Ensures standardized structure for reliable data processing.
- Priority: High

BR-REQ-003 — Computed Column for Enrichment

- Description: Add computed column TotalAmount on order table (PricePerUnit × Qty).
- Business Value: Supports essential analytics and increases data utility.
- Priority: High

BR-REQ-004 — Data Cleansing

- Description: Remove "Null"/Null literals and duplicate records from customer and order tables.
- Business Value: Improves data quality and prevents erroneous analytics.
- Priority: High

BR-REQ-005 — ordersummary Table Creation

- Description: Create new ordersummary table by joining customer and order on CustId.
- Business Value: Centralizes data for downstream reporting and analytics.
- Priority: High

BR-REQ-006 — SCD Type 2 Handling

- Description: Apply SCD Type 2 change handling for customer data in ordersummary, maintaining full history.

- Business Value: Provides auditability and accurate historical analysis.
- Priority: High

BR-REQ-007 — SCD Type 2 Rules

- Description: Set previous customer records to Inactive, new ones to Active, and maintain StartDate/EndDate.
- Business Value: Enables comprehensive tracking of changes over time.
- Priority: High

BR-REQ-008 — Aggregate Table Creation

- Description: Create customeraggregatespend table aggregating spend by Name and Date.
- Business Value: Delivers summarized insights for business analysis.
- Priority: High

BR-REQ-009 — Aggregate Processing

- Description: Aggregate customer spend by grouping Name and Date, summing TotalAmount from ordersummary.
- Business Value: Facilitates snapshot analysis and trend identification.
- Priority: High

BR-REQ-010 — Solution Modality

- Description: Implement all steps above using Delta Live Tables.
- Business Value: Standardizes pipeline execution and management.
- Priority: High

Assumptions & Constraints

- No assumptions beyond explicit JSON content.
- Delta Live Tables must be used for implementation.
- Additional SCD Type 2 attributes (StartDate, EndDate, Active) referenced in requirements but not fully defined.

Success Metrics

- All source data is correctly loaded into Delta tables.
- Data quality is maintained by effective cleansing steps.
- ordersummary accurately reflects SCD Type 2 history for customer data.
- customeraggregatespend table summarizes total spend accurately and matches source records.
- Every pipeline step (1–10) executed successfully via Delta Live Tables.

Risks & Mitigations

- Ambiguity in specific SCD Type 2 attribute fields: Mitigation—document open questions for follow-up.
- Data quality assurance beyond removal of Nulls and duplicates is undefined: Mitigation—flag as out-of-scope.
- Lack of explicit error handling, scheduling, and monitoring: Mitigation—recommend future requirements elaboration.

Approval

- Approver: Not present in source JSON
- Date: Not present in source JSON

Appendix

Appendix A: Source Definitions

Source Name	Type	Location	Format	Notes
Customer source	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata	CSV	Loaded into Delta table customer
Order source	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata	CSV	Loaded into Delta table order

Appendix B: Target Definitions

Layer	Table	Columns	Notes
Raw	customer	CustId, Name, EmailId, Region	Delta table
Raw	order	OrderId, ItemName, PricePerUnit, Qty, Date, CustId	Delta table
Processed	ordersummary	CustId, Name, EmailId, Region, OrderId, ItemName, PricePerUnit, Qty, Date	SCD Type 2 ha
Aggregate	customeraggregatespend	Name, TotalAmount, Date	Aggregated spe

Appendix C: Transformations

Step	Transformation	Input	Output	Logic
1	Load CSV	customerdata, orderdata	customer, order	Copy as per defined schema
2	Remove Nulls	customer, order	customer, order	Filter “Null”/Null from all fields
3	Deduplicate	customer, order	customer, order	Drop duplicate records
4	Add TotalAmount	order	order	PricePerUnit × Qty
5	Join	customer + order	ordersummary	Join on CustId
6	Apply SCD Type 2	customer data in ordersummary	ordersummary	Maintain Inactive/Active, StartDate/EndDate
7	Aggregate	ordersummary	customeraggregatespend	Group by Name and Date, sum TotalAmount

Appendix D: Business Rules

Rule ID	Area	Description	Source
BR-001	Computed Column	TotalAmount = PricePerUnit × Qty	JSON
BR-002	Cleansing	Remove “Null”/Null and duplicates	JSON
BR-003	Join	Join customer to order on CustId	JSON
BR-004	SCD Type 2	Maintain historical records with Inactive/Active, StartDate/EndDate	JSON
BR-005	Aggregation	Group by Name and Date, sum TotalAmount	JSON

Appendix E: Process Flow

Step	Description	Dependency
1	Load customer and order CSVs to Delta tables	Source files available
2	Remove “Null”/Null literals and duplicates	Data loaded
3	Compute TotalAmount in order	Data cleansed
4	Join orders with customers by CustId	Prior steps completed
5	Create ordersummary	Join completed
6	Apply SCD Type 2 handling	ordersummary created
7	Aggregate spend per customer per date	ordersummary available
8	Create customeraggregatespend table	Aggregation completed

Diagram:

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